LEVEL 1 — Fundamentals (Electronics + C)

1. Basic Electronics (non-negotiable)

- NPTEL – Basic Electronics course
- MIT OpenCourseWare – Circuits and Electronics

Focus:

- Ohm’s Law, KCL, KVL
- Diodes, transistors
- Basic circuit analysis

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2. Digital Logic

- NPTEL – Digital Circuits

Focus:

- Logic gates
- Flip-flops
- Counters

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3. C Programming (Embedded-focused)

- freeCodeCamp – C Programming Full Course
- GeeksforGeeks – C Programming Language

Focus:

- pointers
- memory
- bitwise operations

LEVEL 2 — First Hardware Interaction (Arduino)

4. Arduino Basics

- Arduino Official Tutorials

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5. Practice Without Hardware (optional)

- Tinkercad Circuits

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6. Hands-on Learning

- Paul McWhorter Arduino Course (YouTube)

Build:

- LED control
- Button input
- Motor control
- Sensors

LEVEL 3 — Core Embedded Systems (Serious Phase)

7. Embedded Systems Fundamentals

- UT Austin Embedded Systems Shape the World

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8. Embedded C (critical)

- Embedded C Programming Tutorial

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9. Microcontroller Deep Dive (Registers, Peripherals)

- FastBit Embedded Brain Academy (YouTube)

Learn:

- Registers
- GPIO
- Timers
- Interrupts

LEVEL 4 — Advanced Embedded (Industry Skills)

10. Communication Protocols

- I2C, SPI, UART Explained

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11. RTOS (FreeRTOS)

- FreeRTOS Official Documentation

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12. ARM Microcontrollers (STM32)

- STM32 Tutorials (Controllerstech)

LEVEL 5 — Advanced / Near-Mastery

13. Embedded Linux (High-level systems)

- Embedded Linux Primer (free book)

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14. Real Projects + System Design

- Hackster.io (project ideas)

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15. Datasheets Mastery (Real skill)

- AllDatasheet (component datasheets)
